Pave Screen

Pressing the Home Button navigates to the Pave Screen where paving function status is displayed, along with a navigation banner for making speed adjustments. **(Figure 4-29)**



Figure 4-29. Pave Screen

Engine status icons appear across the top of the screen, with auger, conveyor, DPF level, generator, and screed temperature shown in the row beneath.

NOTE: If "fault" appears instead of screed temperature, there is an issue with the sensor.

The auger, conveyor and generator icons are the same symbols shown on the main control panel and screed keypad (generator is not shown on the screed keypad). Conveyor speed is shown just below its icon. The auger and conveyor icons display in three color states:

Red: Off

Green: On

Blue: Paused.

When you turn on the generator to heat the screed, the generator icon changes from red (OFF) to green (ON). If operating in MANUAL mode, the actual temperature is shown in black. If operating generator in AUTO mode, the actual temperature is shown in black.

NOTE: The IDLE throttle speed button ceases to function when the generator is on. When the generator turns off, the IDLE functionality returns.

NOTE: The generator is set at the factory to default at 1800 RPMs.

Important gauges are shown beneath the row of paving status icons. **(Figure 4-30)**

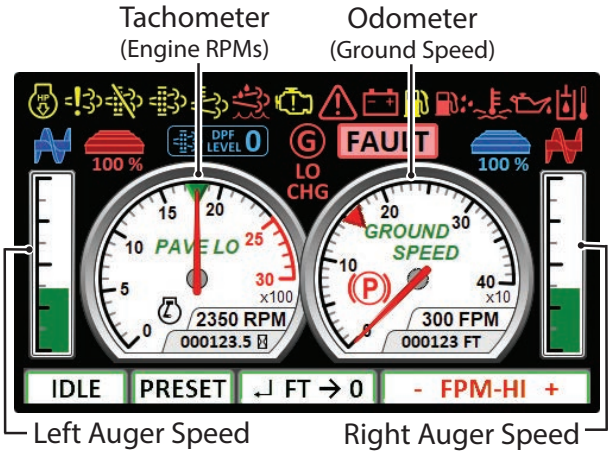


Figure 4-30. Pave Screen Gauges

- Bar graphs on the left and right sides of the screen reflect auger speed (set by the operator on the auger dial if operating in the MANUAL mode).
- The tachometer shows engine RPMs (same gauge as on the engine screen). The red needle indicates the actual speed. The set point triangle indicates speed set points. "Pave Lo" or "Pave Hi" appears in the tachometer denoting whether operating in low or high RPMs.
- The set point triangle automatically changes color when shifting between low gear [green] or high gear [red]. "FPM-HI" [high gear] or "FPM" [low gear] also displays on the FPM (feet per minute) navigation banner above Button #4.

- The ground speed gauge displays a digital value for actual speed and odometer that accumulates distance in feet. The red needle indicates the actual ground speed. The green triangle indicates speed set points.
- The maximum ground speed can be adjusted using Buttons #3 (-) [decrease] and #4 (+) [increase]. **(Figure 4-31)**
- Pressing and holding the center of the navigation button (↵) resets the odometer to zero feet.

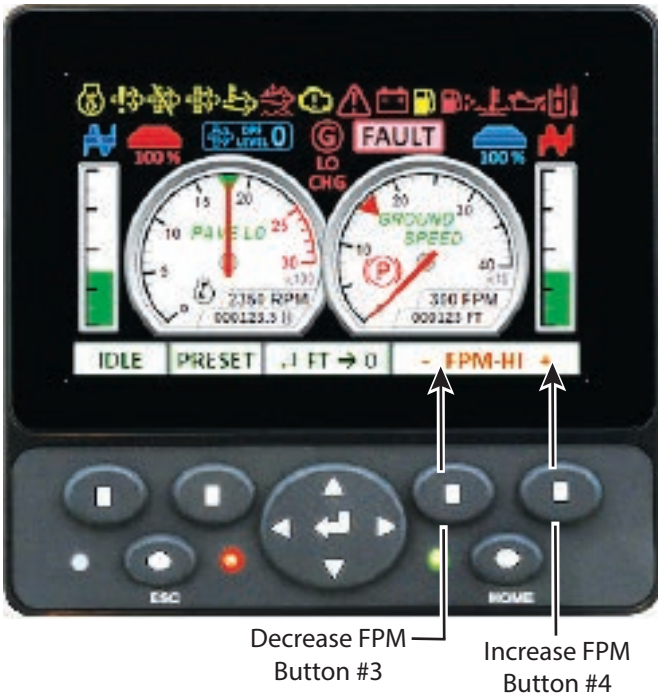


Figure 4-31. Ground Speed Adjustment

Pave Screen Alert Messages

If a fault is present, a red warning icon ⚠ will flash (on both the engine and pave screens). If the icon is solid, the main controller is offline and cannot communicate with the display, therefore no fault information can be shown. A list of paving-related fault codes is shown on the following pages.

You can view active **(Figure 4-32)** and logged **(Figure 33)** system faults by pressing the screen navigation left or right from any screen.



Figure 4-32. Active Paving Function Faults Screen

- Press Button #1 to view active paving function faults.
- Press Button #2 to view logged paving function faults.
- Use Buttons #3 and #4 (PREV [previous] and NEXT) to scroll between fault screens.



Figure 4-33. Logged Paving Function Faults Screen

Pop-Up message banners may also appear on the pave (and engine) screen to alert the operator of an issue, or if the operator attempts to do something under the wrong conditions. **(Figures 4-34 to 4-42)**

The following banner appears if the operator attempts to crank the engine with the auger, conveyor or generator are engaged. **(Figure 4-34)** This banner notifies the operator that the engine will not start unless these components are turned off.

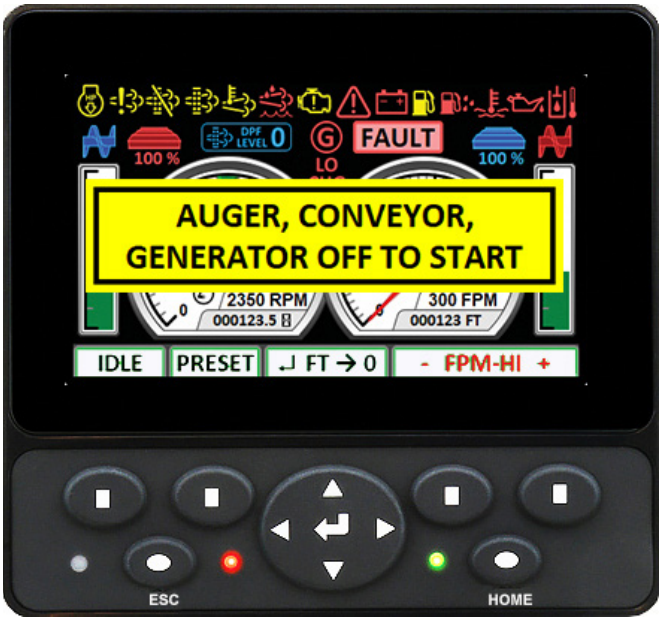


Figure 4-34. Component Engine Start Alert

The following pop-up messages appears if the operator attempts to activate a control panel while parked regeneration is in process. **(Figure 4-35)**



Figure 4-35. Parked Regen Active Alert

If the operator attempts to crank the engine or activate a control panel console while one or more joystick(s) is out of neutral, an alert message appears. **(Figure 4-36)**

NOTE: The joystick in the wrong position will be shown in red.



Figure 4-36. Joysticks Must Be in Neutral Alert

The following message alert appears if the paver in high gear with the conveyors or augers on. **(Figure 4-37)**



Figure 4-37. High Gear Pave Alert

If the operator attempts to shift from high to low gear while traveling too fast, the following message pops-up on the screen. **(Figure 4-38)**

NOTE: The downshift will automatically occur if the operator slows down.



Figure 4-38. Downshift Inhibited Alert

If the operator is attempting to use the joystick(s) on an inactive console, the following message appears that notifies operator to activate the control panel console. **(Figure 4-39)**



Figure 4-39. Activate Console Alert

If the control panel or a joystick is disabled or faulty, the following message appears: **(Figure 4-40)**



Figure 4-40. Joysticks Must Be in Neutral Alert

The following message alert appears if the operator sets the RPMs above the Pave Hi (high) preset. **(Figure 4-41)**



Figure 4-41. High Engine RPM Alert

The following messages appears if the RPM is set above idle while turbo protection is active. **(Figure 4-42)**

NOTE: The turbo protection feature takes control of the throttle for the first 1 - 30 seconds based on ambient temperatures. It does not allow the engine RPM to increase to ensure proper lubrication to the turbo and cylinders before increasing RPMs.



Figure 4-42. Throttle Up Inhibited Alert

Pave Function Fault Codes

Fault Code	Fault Description
1	Left Display Time Out
2	Left Keypad (Lower Left) Time Out
3	Left Keypad (Lower Right) Time Out
4	Left Keypad (Upper Left) Time Out
5	Left Keypad Time Out
6	Left Keypad (Upper Right) Time Out
7	Left Console, Left Joystick Fault
8	Left Console, Left Neutral Fault
9	Left Console, Right Joystick/Steer Fault
10	Left Console, Right Neutral Fault
11	Left Console, Auger Potentiometer Fault
12	Left Screed, Auger Potentiometer Fault
17	Right Display Time Out
18	Right Keypad (Upper Left) Time Out
19	Right Keypad (Lower Left) Time Out
20	Right Keypad (Upper Right) Time Out
21	Right Keypad (Lower Right) Time Out
22	Right Keypad Time Out
23	Right Console Left Joystick Fault
24	Right Console Left Neutral Fault
25	Right Console Right Joystick/Steer Fault
26	Right Console Right Neutral Fault
27	Right Console Auger Potentiometer Fault
28	Right Screed Auger Potentiometer Fault
33	Safety Controller Error Detected
34	Engine J1939 Time out
35	Engine ACU J939 Time Out
36	Engine ACU2 J1939 Time Out
37	Input Fault, Fuel Level Sender
38	Press Sensor 1 Fault, Auger/Work Pump
39	Press Sensor 2 Fault, Conveyor Manifold
40	Press Sensor 3 Fault, Cylinder Manifold
41	Press Sensor 4 Fault, Propel Left Forward
42	Press Sensor 5 Fault, Propel Right Forward
43	Sensor Power Fault
44	Input Fault, Left Auger Sonic
45	Input Fault, Right Auger Sonic
46	Input Fault, Left Propel Speed Sensor

Fault Code	Fault Description
47	Input Fault, Right Propel Speed Sensor
48	Input Fault, Hydraulic Temperature Sensor
49	Input Fault, Screed Temperature Sensor
50	Cylinder Output Module Time Out
51	Motor Output Module Time Out
52	Screed Output Module Time Out
53	Output Fault, Left Auger Proportional
54	Output Fault, Left Auger Reverser
55	Output Fault, Right Auger Proportional
56	Output Fault, Right Auger Reverser
57	Output Fault, Beacon Light
58	Output Fault, Left Berm Down
59	Output Fault, Left Berm Up
60	Output Fault, Right Berm Down
61	Output Fault, Right Berm Up
62	Output Fault, Brake Release
63	Output Fault, Backup Alarm
64	Output Fault, Left Conveyor Proportional
65	Output Fault, Left Conveyor Reverser
66	Output Fault, Right Conveyor Proportional
67	Output Fault, Right Conveyor Reverser
68	Output Fault, Left Cut-Off Close
69	Output Fault, Left Cut-Off Open
70	Output Fault, Right Cut-Off Close
71	Output Fault, Right Cut-Off Open
72	Output Fault, Engine On
73	Output Fault, Engine Start
74	Output Fault, Left Extension
75	Output Fault, Right Extension
76	Output Fault, Generator Neutral Signal
77	Output Fault, Generator On
78	Output Fault, Truck Hitch In
79	Output Fault, Truck Hitch Out
80	Output Fault, Left Hopper Close
81	Output Fault, Left Hopper Open

Fault Code	Fault Description
82	Output Fault, Right Hopper Close
83	Output Fault, Right Hopper Open
84	Output Fault, Horn
85	Output Fault, Lights
86	Output Fault, Propel High Gear
87	Output Fault, Left Propel Forward
88	Output Fault, Left Propel Reverse
89	Output Fault, Right Propel Forward
90	Output Fault, Right Propel Reverse
91	Output Fault, Screed Down
92	Output Fault, Screed Up
93	Output Fault, Left Slope Down
94	Output Fault, Left Slope Up
95	Output Fault, Right Slope Down
96	Output Fault, Right Slope Up
97	Output Fault, Left Sonic Enable
98	Output Fault, Right Sonic Enable
99	Output Fault, Spray Down Pump
100	Output Fault, Left Tow Down
101	Output Fault, Left Tow Up
102	Output Fault, Right Tow Down
103	Output Fault, Right Tow Up
104	Output Fault, Vibrator
105	Engine Shut Down, Low Oil Pressure
106	Engine Shut Down, High Coolant Temperature
107	Propel Pump Charge Pressure Low
108	Engine Oil Pressure Low
109	Water in Fuel
110	Engine Coolant High Temperature
111	Hydraulic Oil Temperature
112	Primary/Secondary Processor Mismatch
113	No Valid Parameter File Loaded
114	Output Fault, Track Tension Unloader

Operator Settings Screen

Navigate to the Operator Settings screens to set paving controls by pressing left or right on the screen navigation button. **(Figures 4-43 and 4-44)** Use this screen to select manual or AUTO mode for operating the under auger material cut-offs, conveyor and generator. You can set the screed temperature for automatic temperature control or choose to operate the screed heat manually using the keypad. You can also disable one or both screed control panels.

- Use the up and down arrow buttons to scroll to each setting window. Pressing the left/right arrow (right) to the end will navigate to the second operations settings screen.
- Press the center of the navigation button (↵) to select desired setting when highlighted in the window.

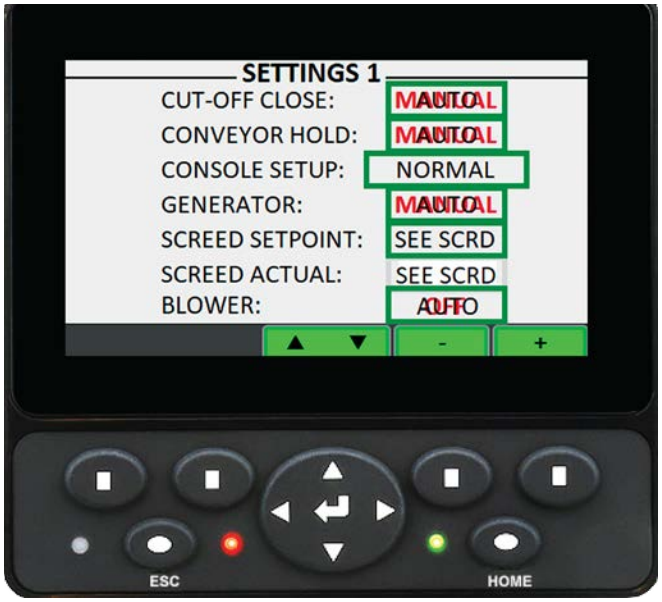


Figure 4-43. Operator Settings Initial Screen

Cut-Off Close: There are two settings for the under auger material cut-offs: Manual and AUTO.

- In AUTO mode, the cut-off will automatically close when putting the paver in reverse.

Conveyor Hold: There are two settings for the conveyor hold: Manual and AUTO.

- In AUTO mode, the conveyors will stop when the paver stops.

Console Setup: You can disable the left, right or both main control panels in this window in the event of fault conditions. For example, if joysticks are malfunctioning on one console, disable that console and use the other side.

Generator: Set the generator to operate in manual or AUTO mode in this window.

- In auto, the generator will turn on and off automatically to maintain the temperature set point.
- In manual mode, the generator will turn on and off after an elapsed timer.

NOTE: If the temperature sensor is faulted, manual mode will be forced.

Screed Set Point: Use this window to adjust the screed set point temperature.

- Use the plus and minus buttons for setting the screed set point temperature.
- If in manual mode, N/A will be shown in the window. This is the default setting.

Screed Actual: This is the actual measured screed temperature.

NOTE: If a fault exists in the sensor, FAULT will be shown rather than the temperature.

Blower: The blower will automatically turn on when either conveyor is turned on in AUTO mode, and shut off when conveyors are both off.

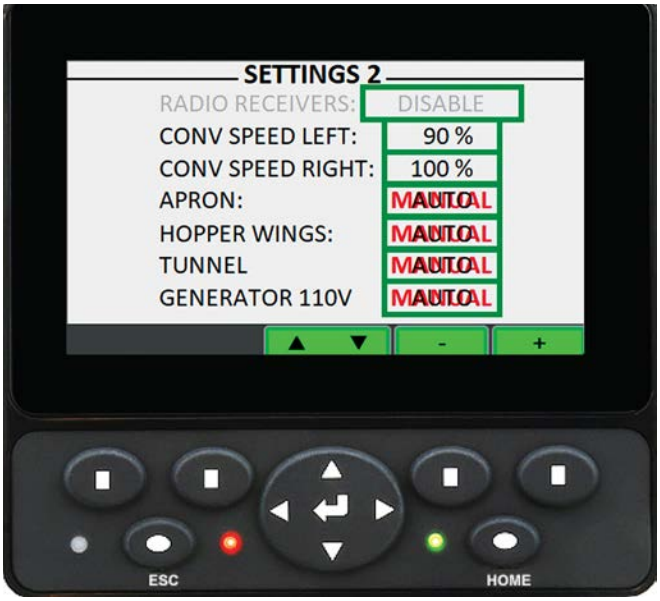


Figure 4-44. Operator Settings Secondary Screen

Conveyor Speed Left/Right: Sets the maximum speed allowed by the conveyor paddles.

Apron: When set to AUTO, the system will automatically raise/lower the apron when the hopper is opened and closed.

Hopper Wings: When set to AUTO, the system will automatically run with the hopper open when the operator presses the Hopper Open button.

Tunnel: When set to AUTO, the tunnel will automatically function while operating the screed extension.

Generator: In AUTO mode, the generator turns on and off automatically to maintain the temperature set point. In MANUAL mode, the generator turns on and off on a timer basis.

NOTE: If the temperature sensor fails, manual mode will be forced.

Motor Settings Screen

Navigate to the Motor Settings screen by the pressing left or right on the screen navigation button. **(Figure 4-45)**

Use this screen to set the left/right augers and conveyor for operating in forward or reverse. You can also set the left and right conveyor speed on this screen.

- Use the up and down arrow buttons to scroll to each setting window.
- Press the center of the navigation button (↵) to select desired setting when highlighted in the window.
- Use the plus and minus buttons for setting the conveyor speed.



Figure 4-45. Motor Settings Screen

Pressure Gauge Screen

Use the Home button to cycle between the Pave screen and the Pressure Gauge screen. The pressure gauge screen displays hydraulic pressure information for five paver functions: auger, conveyor, cylinders, propel left and propel right. **(Figure 4-46)**

NOTE: If a fault exists, the LO CHG icon will illuminate, and the pressure digital read-out background for that component will turn red.

- Press Button #1 to idle the engine.
- Press Button #4 to use preset RPMs, or adjust engine RPMs:
  - Press Button #2 to decrease RPMs.
  - Press Button #3 to increase RPMs.

NOTE: The engine RPMs is shown in the center of the navigation banner.

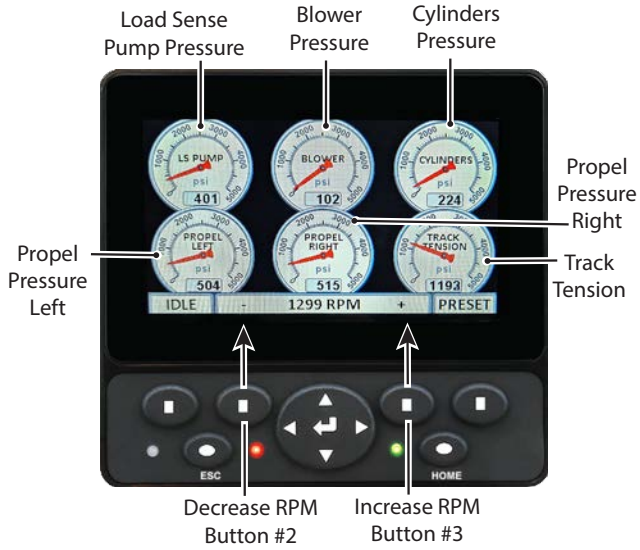


Figure 4-46. Pressure Gauge Screen

DPF and Regeneration

It cannot be emphasized enough that the paver should be maintained in good operating condition so the emission system can automatically perform the regeneration (regen) process.

⚠CAUTION If engine faults are present that prevent the ability to perform regeneration, DPF levels will continue to rise. It is important to resolve any engine faults immediately and perform the needed regeneration to prevent costly DPF technician service if DPF level warnings or engine faults are ignored.

If regeneration begins while operating, immediately move the paver to a safe location to allow regeneration to occur. During regeneration, there may be a noticeable sound difference. The high exhaust temperature icon will illuminate on engine and pave screens will appear during the process.

- If unable to move to a safe location for the engine to perform a regeneration automatically, the operator may choose to ***temporarily*** inhibit (disable) the regeneration process.
  - Press Button #4 while on the Engine screen, then use Button #2 to inhibit regeneration, described on ***Page 4-27***, until you can move to a safe location and complete the parked regeneration process. ***(Page 4-44)***

NOTE: The ability to inhibit (disable) regeneration is only possible at DPF levels 1 and 2, described on the following page.

NOTE: Regeneration will remain disabled until the engine is turned off or the operator navigates back to the screen to enable regeneration.

⚠CAUTION EXHAUST TEMPERATURES WILL BE EXTREMELY HIGH. BE SURE EXHAUST WILL NOT COME INTO CONTACT WITH ANY COMBUSTIBLE MATERIALS.

⚠CAUTION Monitor the surrounding area during regeneration. If unsafe conditions develop, shut the unit off immediately.

DPF Levels

Pay attention to the DPF level icon on the engine and pave screens. Automatic regeneration can only occur at DPF Level 1 and 2.

DO NOT allow soot levels to reach high levels to prevent damage to the engine and/or DPF components that will require costly repairs. DPF levels are described below.

NOTE: If the operator has not inhibited regeneration and there are no engine faults, but automatic regeneration is not occurring at DPF Level 1 or 2, increase engine RPMs and/or load to allow the DPF to automatically regenerate.

NOTICE **It is important to perform a parked regeneration BEFORE the DPF level reaches Level 4. If the regeneration process is not performed and the soot level reaches Level 4, non-warranty factory assistance from the engine manufacturer is required.**

If an engine fault is present at DPF Levels 1 and 2, a message banner may appear on the engine and pave screen notifying operator regeneration is not possible until the engine fault is resolved. DPF levels will continue to rise. **(Figure 4-47)**



Figure 4-47. Active Engine Code DPF Alert

DPF Level 1: The DPF is loaded with soot and needs to be regenerated. When safe, please uninhibit automatic active regeneration.

DPF Level 2: The soot level has continued to accumulate. Increase engine RPMs (if at idle) and engine load to prompt automatic regeneration.

NOTE: If the operator is inhibiting automatic regeneration and does not perform parked regeneration, the soot level will continue to rise.

DPF Level 3: An orange message alert banner may appear to warn the operator that soot levels are extremely high. **(Figure 4-48)** If you have allowed the exhaust soot level to reach Level 3, automatic regeneration is not possible. If this message appears, the operator must perform parked regeneration at the earliest opportunity. When safe, complete a parked regeneration as described on the following pages.

NOTICE **Continued operation without completing a parked regeneration can cause engine damage and will void engine warranty. The operator should perform parked regeneration immediately!**



Figure 4-48. Parked Regeneration Required Alert

DPF Level 4: If the operator continues ignoring DPF level warnings and the soot level reaches Level 4, a red message alert banner appears on the engine and pave screens. **(Figure 4-49)**

Engine output is limited and the operator can no longer perform regeneration. Regeneration is required IMMEDIATELY by a qualified engine or OEM service technician using specialized equipment. Move the paver to a safe location, shut down, and contact your authorized dealer for service.

NOTICE **If the soot level reaches Level 4 or 5, specialized equipment is needed to perform regeneration and the machine must be serviced by your authorized dealer. The DPF system may become damaged and require replacement. Therefore, performing regeneration BEFORE reaching Level 4 is imperative for preventing costly repair.**



Figure 4-49. DPF Exceeds Normal Limits Alert

Regen Level 5: If the DPF level reaches Level 5, a red message alert banner appears on the engine and pave screens notifying operator the DPF must be replaced. **(Figure 4-50)** The DPF must be removed and cleaned by a qualified cleaning facility, and the ECU must be reset by a qualified engine or OEM service technician. The engine will derate to idle with no torque available. IMMEDIATELY contact your authorized dealer.



Figure 4-50. DPF Must Be Replaced Alert

Parked Regeneration

If the DPF is at Level 2 or 3, and no engine faults are present, perform parked regeneration using the following procedure:

1. Park the paver in an open area away from people, equipment, trees/shrubbery, debris, and combustible materials as exhaust temperatures will increase considerably during regeneration.
2. Place the paver in NEUTRAL and engage the park brake.
   - Ensure the coolant temperature is the operating range of 155° to 160° F.
3. Set the engine to low idle.

⚠CAUTION DO NOT ATTEMPT TO OPERATE THE UNIT, CHANGE ENGINE RPM OR MOVE FROM PARK/NEUTRAL WHILE REGEN IS OCCURRING. This will abort the Regeneration process and require you to start the process over.

4. From the engine screen, press Button #4 to access DPF commands. A pop-up banner appears with regeneration selections corresponding to the buttons. **(Figure 4-51)**

NOTE: If the icon background is gray, the function is unavailable (or unneeded) at the moment.

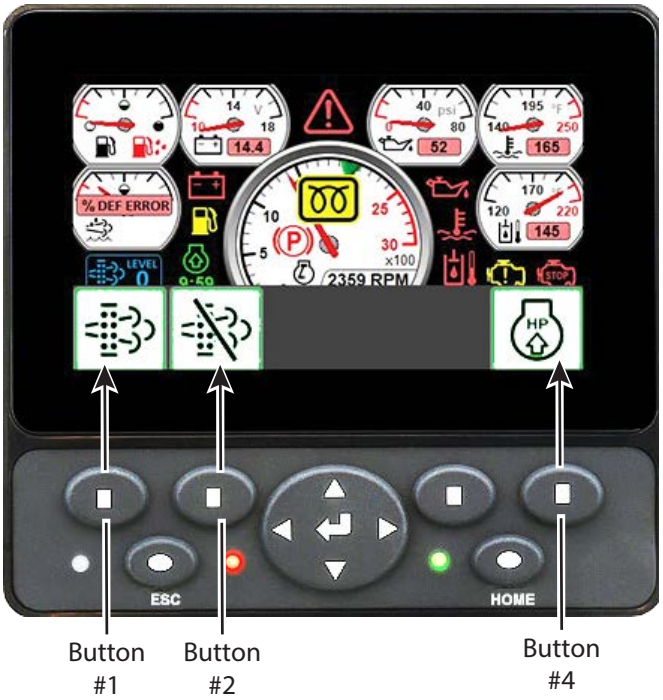


Figure 4-51. Engine Screen Pop-Up Banner

5. Press Button #1 to start the regeneration process.

During this process:

- DO NOT move machine out of park or neutral.
- DO NOT touch the accelerator pedal/lever.
- DO NOT turn the engine off or stop the regeneration process.

- The engine speed will increase and there may be a noticeable sound difference. This is normal. The "Parked DPF Regen Active" message alert will appear **(Figure 4-52)**.

NOTE: The High Exhaust icon will illuminate when the exhaust temperature reaches >842° F Both alerts will stay on during the regeneration process.



Figure 4-52. Parked Regen Active Alert

6. Monitor your surroundings while the system is regenerating, ensuring others stay away from the machine. If any unsafe condition arises, shut off the engine immediately and restart the process when the area is clear.

When the parked regeneration process is complete, the engine will automatically return to normal idle speed. The Parked DPF Regen Active message will disappear and the high exhaust icon light will turn off.

PLUS +1® CONTROLLERS

The Plus +1® system contains several controllers and OX-modules operating in the background that monitors paver control systems and alerts the operator in the event of a fault. This is the brains behind paver and engine systems. When it senses a fault, the Plus +1® sends a message to the DM430 display to alert the operator. The Plus +1 controller and OX-modules are located throughout the engine compartment and on the screed. (A complete list of fault codes with corrective action messages are shown in **Section 5,** *Troubleshooting*.**)**

The Plus One system will not allow the engine to start if:

- Steering control box not detected.
- Joystick(s) not in neutral at start-up.
- Pump control coil fault.
- Brake valve/pump neutral bypass coil fault.



DRIVING THE 8680 PAVER

The dual operator platforms allow easy and convenient access for controls of most paver functions. The paver is usually operated from the left operator station, but can also be operated from the right side. Dual joysticks propel forward and reverse, and steer the paver. Ramping for acceleration and deceleration are proportional to the movement of the joysticks. The amount of steering is controlled by the amount of joystick movement and the speed of the paver.

⚠ DANGER NEVER leave the operator platform unattended while the paver is in gear or in motion. Operator must return joystick(s) to the neutral position, engage the park brake, press the Start/Stop button, and remove the ignition key before leaving the operator platform.

⚠ WARNING Ensure there are no people, equipment or obstacles in the paver's before moving in either direction.

Make seat and operator platform adjustments described on ***Page 4-5*** before operating the paver. Follow the instructions on ***Page 4-6*** to start the engine and activate the desired operator main control panel.

4

Keep these items in mind for operating with rubber tracks and maximizing track life:

- Avoid detracking episodes by:
  - Maneuvering at lowest speed possible;
  - Avoiding sharp turns on hot asphalt (move to a cold area for turns);
  - Making gradual turns in as large an arc as possible, repeating zigzag motions gradually instead of pivoting;
  - Using forward instead of reverse when possible;
  - Preventing large foreign objects from being entangled in the undercarriage/tracks.
- Properly trained operators who practice good operating techniques will make a big difference.
- Avoid spot turns, spinning, constant operation on slopes and contact with the side of tracks on curbs, hard walls and other objects. Limit use around large, sharp and rocky surfaces such as manholes and drainage covers.
- Clean, inspect and check alignment on the rubber tracks at least once per day. Replace worn parts immediately if needed.
- Avoid track contact with abrasive/corrosive materials. If oil or similar products get on the tracks, clean it off immediately. Over time, oil degrades rubber quality.
- Remove asphalt buildup on the undercarriage components and rubber tracks. Buildup causes misalignment between the drive wheel and guide lugs, poor tracking performance and premature undercarriage components/rubber track wear. Periodically check the undercarriage components for wear (i.e., drive wheel, rollers and idler) and make repairs if needed.



Steering and Speed Control

Use the dual joysticks for steering and speed control. Move the joysticks SLOWLY. Lift up on the joystick collar and move joystick either FORWARD or REVERSE. The joystick collar will lock back into place in the NEUTRAL position. **(Figure 4-53)**

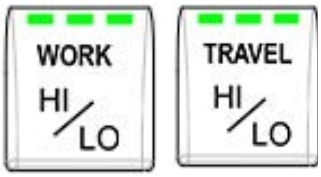
⚠ DANGER Rapid counter-rotation in either direction can result in the operator being thrown from the machine and suffer personal injury or death. Counter-rotation occurs if one joystick is in the extreme forward position and the other is in the extreme rear position while the machine is in gear and moving. Make only SLOW adjustments to the joysticks to turn the paver right or left.

⚠ CAUTION Be aware of the terrain, traffic and other driving conditions. Adjust speed accordingly.

⚠ CAUTION Avoid low limbs, power lines, and other objects that can endanger crew or paver.

NOTICE **ALWAYS pave in Work mode. Use Travel mode only for traveling.**

To drive the 8680 Commercial Paver:



Select WORK Hi/Lo or TRAVEL Hi/Lo mode on the main control panel, then use joysticks to move forward and reverse.



Figure 4-53. Joysticks

- To slow the paver, move joysticks closer to the neutral position.
- To stop the paver, pull joysticks back to the neutral position.
- To drive the paver forward, lift up on the joystick collar and push joystick forward to reach the desired speed.
- To move in reverse, lift up on the joystick collar and move joystick backwards to reach desired speed.
- Return joysticks to the neutral position to stop paver.
- To steer the unit to the left, SLOWLY push the right joystick farther forward than the left joystick. The farther the joystick is pushed, the more the paver turns. Slow and easy adjustments are required for safe operation.
- To steer the unit to the right, SLOWLY push the left joystick farther forward than the right joystick. The farther the joystick is pushed, the more the paver turns. Slow and easy adjustments are required for safe operation.

If you need to stop the paver:

1. Place joysticks into the neutral position.
2. Throttle back to idle using settings on the DM430 digital display screen.
3. Turn ignition key on the junction box counterclockwise to the OFF position and remove key.

NOTE: If the engine does not shut down when the key is turned to OFF, press the E-Stop button on the control panel to terminate power.



You can also use the Pause button once to pause propel, conveyors and augers functions. Pressing the button again resumes these functions.

NOTE: The engine idle will remain at high but automatically go into low idle if paused for a preset period of time (unless the generator is enabled.)

If equipped with the optional steering knob, it is used for steering and replaces the right joystick on the control panel. **(Figure 4-54)** Use the left joystick for forward and reverse speed control and steer right or left using the steering knob.

1 - Press button for incremental steering to the left.

2 - Press button to center paver.

3 - Press button for incremental steering to the right.

4 - Turn knob left to steer left; Turn knob right to steer right.

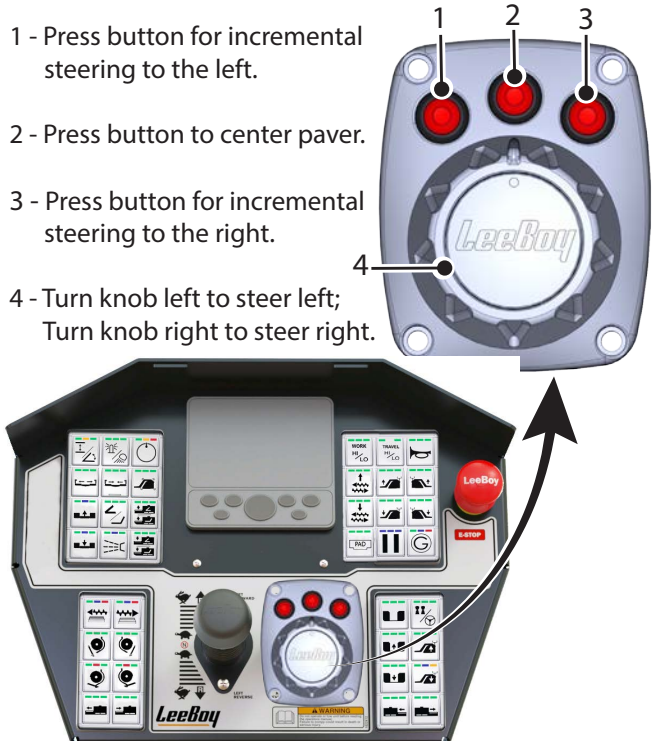


Figure 4-54. Steering Knob

4

If equipped with the optional steering wheel, it is used for steering and replaces the right joystick on the control panel. **(Figure 4-55)**

The joystick is used for speed control and propelling forward and in reverse. The center position is neutral.

- To drive the paver forward, lift up on the joystick collar and push joystick forward to reach the desired speed.
- To move in reverse, lift up on the joystick collar and move joystick backwards to reach desired speed.
- Return joystick to the neutral position to stop paver.



Figure 4-55. Steering Wheel Option



PAVER OPERATION

Spray Down

Always spray down the machine to clean it before and after paving. Spray solvent or release agent on any part of the paver that comes into contact with asphalt to prevent buildup that will damage the machine over time.

The citrus tank with filler cap is located under the right floorboard beside of the DEF tank. **(Figure 4-56)**



Figure 4-56. DEF and Citrus Fill Caps

⚠CAUTION Ensure the area is clear of others and flammable materials before spraying. Allow the screed to cool at least 10 minutes before spraying to prevent possible burns from solvent spray on a hot screed. The screed should be warm, but not hot.

⚠WARNING Never spray cleaning solvent or release agent near the screed heating element or near open flame. These agents are highly flammable and can ignite, causing serious personal injury.

NOTICE **When using solvents and release agents, DO NOT allow cleaning solvent to run onto the ground. Only spray down the machine in a designated area away from the paving site.**

There is a spray down system on both sides of the Leeboy 8680 paver. A spray wand is conveniently located on each side of the operator platform. **(Figure 4-57)** The hose reels are located inside the access panel on each side of the paver.



Figure 4-57. Spray Down Wand Location (Each Side)

To operate the spray down system:



1. Push the Spray Down button on the main control panel to turn system ON.
   - Press the button again to turn OFF.
2. Extend hose from the hose reel.
3. Squeeze the wand handle to spray desired areas.

Start at the front of the machine and work rearward, using a scraper if needed, paying particular attention to:

- Push Rollers: Rotate the rollers while cleaning and soak bearing ends to maintain free rotation.
- Rubber Tracks and Undercarriage: The importance of keeping the heavy-duty rubber tracks cannot be emphasized enough to prevent premature wear. Don't operate in corrosive, abrasive, salt or other contaminant materials that cause rubber to deteriorate. Look for oil and grease that may drip from the machine onto the tracks and rinse off as soon as possible. Mud can also harden and adversely affect track operation along with other problems. You can use a pressure washer to remove the mud. Cleaning is especially important in colder temperatures.

NOTE: DO NOT operate with frozen tracks. If you try to use power to force the tracks to move, you could damage them or your final drive.

- Hopper and Conveyors: Thoroughly clean surfaces and any material accumulating in chains, drive sprockets, idler shaft, bearings and tunnel areas. AFTER cleaning the machine, operate the conveyors briefly to carry solvent into return areas where it can help dissolve hidden asphalt material.
- Augers: Scrape away any accumulated material around bearings and clean contact surfaces down to the metal to avoid quick build-up while paving.
- Screed: Face and bottom plates should be clean and smooth on the main and screed extensions to achieve a smooth paving mat.
- Operator Platforms: Be sure to wash away solvent residue so walk boards and operator platforms do not become slippery.



Hopper Wings

The hopper wings are hydraulically opened so the hopper can be filled with paving material. The 8680 paver features a 9-ton, rounded design to reduce cold material build-up while rear hopper wing wipers help prevent spillage. Material in the hopper is moved toward the back of the paver to the screed by conveyors.

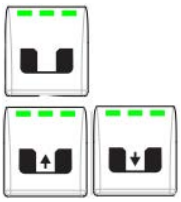
The operator can also extend and retract each hopper wing individually. This feature gives you the ability to pave very close to walls, fences, road barriers, buildings or other obstacles with ease.

⚠WARNING **DO NOT retract the hopper wings until the material truck has moved completely away from the paver or if the hopper is full of asphalt. NEVER raise the conveyor bed with asphalt in the hopper.**

⚠WARNING **Ensure no people, equipment or other obstacles are in or near the hopper when raising or lowering the hopper wings to prevent injury or damage to the paver.**

⚠CAUTION **Ensure the hopper cylinder is unlocked before extending or retracting the hopper wings.**

The hopper wings are operated using hopper buttons on the lower right control panel keypad:

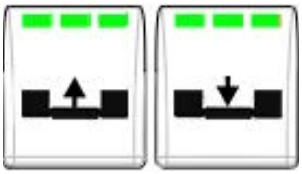


1. Use the hopper wing select button to choose hopper wing function. If operating both hopper wings, there is no need to press the hopper select button as operating both wings is the default operational mode.

- Press the hopper wing select button once to operate the LEFT hopper wing.
- Press the wing selection button again to operate the RIGHT hopper wing.
- Pressing the button again returns to the default operation control for both hopper wings.

⚠CAUTION **Be sure the hopper wings are completely opened before receiving paving material from the truck.**

2. After selecting the left, right or both wings on the hopper wing select button, use the hopper UP and Down buttons to open and close the hopper wings:
   - Press and hold the hopper wing up (Close) or down (Open) buttons to close or open one or both hopper wings based upon the hopper wing selection made.



Press and hold the Apron UP / DOWN button to pivot the front apron to assist in cleaning out the hopper. All three LED lights illuminate while in operation.

Hopper Lock Valves

The hopper cylinder valves can be used to lock each hopper wing when they are open or closed. The lock valves are located inside the left engine panel. **(Figure 4-58)** Be sure to unlock the hopper valves before operation!

- Turn the valve counterclockwise to unlock the hopper wings.
- Turn the valve clockwise to lock the hopper wings.

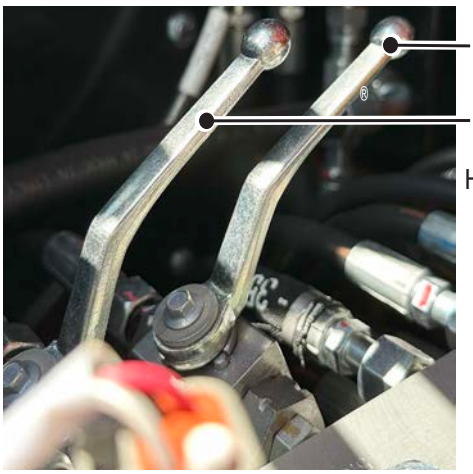


Figure 4-58. Hopper Cylinder Lock Valve



Conveyors

The conveyor is a very important part of the paver that moves the asphalt mix from the hopper to the auger, which controls the amount of material delivered to the screed. Pay close attention to integrating conveyor operation into the total operation of the paver. The left and right conveyors can be operated in conjunction with the augers from either control panel, and both can be operated in reverse.

⚠CAUTION NEVER leave the paver unattended while the conveyor is operating. DO NOT leave paver with empty conveyors running for any length of time to prevent excessive wear.

When operating the conveyor:

- Spray down the conveyor drive chains with cleaning solvent or release agent several times during the work day for optimum performance. ***(See* Section 5 *Lubrication Chart.)***

NOTICE Asphalt in the conveyors can fill tracks and cause damage to the machine. To prevent flight chains from sticking inside the conveyor pans, lubricate them daily for maximum operational efficiency.

- If equipped with the Under Auger Material Cut-Offs option, some spillage can occur while the conveyors are operating. This is normal. To help prevent this, operate conveyors manually when loading hopper and not paving.

- Irregular movement of the conveyor indicates that the conveyor chains need adjustment. The conveyor chains are easily adjusted on the 8680 paver. Fine tuning conveyor chain tension is described in detail in **Section 5.**

NOTE: Check conveyor chain adjustment every 100 hours.

NOTE: Moving the joysticks into neutral will pause conveyor operation. (This feature can be turned off.)

To operate the conveyor(s) from the main control panel:



- Press one or both auger conveyor buttons on either control panel to turn on the conveyor(s) and auger(s).
- AUTO mode is the default. The conveyor automatically stops when proximity sensors detect asphalt height at the appropriate level for pausing the augers conveyors.
- Press and hold the button(s) again for manual operation.

Screed operators can also increase or decrease conveyor speed (and auger speed/pile height) using buttons on the screed control box. **(Figure 4-59)**

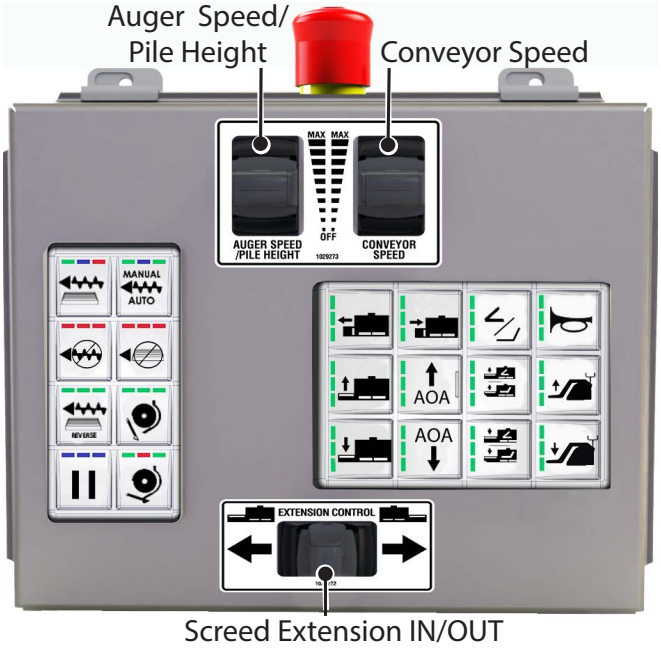
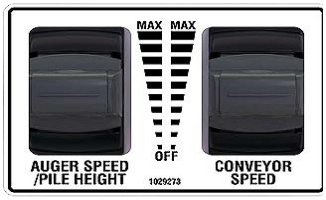
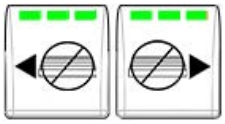


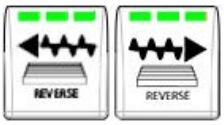
Figure 4-59. Screed Control Box (Left Box Shown)



- Press and hold the button in the desired increase/decrease direction to set conveyor speed. You can also fine tune the conveyor speed in the digital display. ***(See Motor Settings on Page 4-40)***



The screed operator can press the Conveyor Bypass button on the screed control box to pause conveyor operation while paving.



The screed operator can press the Auger/Conveyor Reverse button on the screed control box for reversing the direction of both functions to help clear material as needed.

NOTE: The conveyor will only reverse in the manual mode, and conveyor speed is reduced.

PRO TIP: Use the Pad Builder function on the main control panel to build a starter pad:



- Press the button once to engage the augers and conveyors. Both Under Auger Material Cut-Offs will open.
- Press the button again and the augers/conveyors will turn off.
- Press and hold button to manually engage both conveyors only.



Augers and Sonic Auger Sensor

Independent right and left augers can be operated in manual or AUTO (automatic) mode. The left and right augers are operated in conjunction with the conveyors from either control panel, and both can be operated in reverse.



Press one or both auger/ conveyor button on either the main control panel or the screed control box to turn on the conveyor(s) and auger(s).



- The screed operator can also select manual or AUTO mode after the main operator has turned on the auger/ conveyor using the Auger Manual AUTO button on the screed control box. (AUTO operation is the default.)

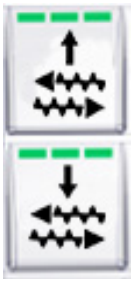
AUTO Mode: The TopCon sonic auger sensor detects the amount of material present and controls its flow to the end gates, gauging the amount of paving material in the extensions.

The sensor sits in its mount on the end gate(s) and connects to its cable on the screed. **The sonic auger sensor system must be connected to operate in AUTO mode. (Figure 4-60)**

- When the material height moves into the sonic range, the system slows the auger speed and turns it off as needed. It will turn back on when the material level drops and continue monitoring pile height.



Figure 4-60. Sonic Auger Sensor



To raise the augers:

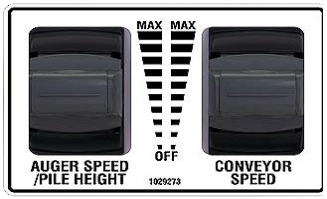
- Press and hold the auger lift button (⇧) to raise both the left and right auger.

To lower the augers:

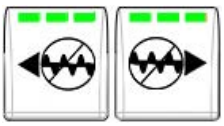
- Press and hold the auger lower button (⇩) to lower both the left and right auger.
- Releasing the button will hold the augers at that position.

NOTE: ALWAYS raise the augers when loading the paver onto transport vehicles.

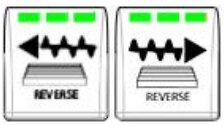
The screed operator can use the Auger Speed / Pile Height button during operation.



- When operating in AUTO mode, press and hold button in the desired increase/decrease direction to set the pile height at the end gate.
- When operating in MANUAL mode, press and hold button in the desired increase/decrease direction to set the auger speed.

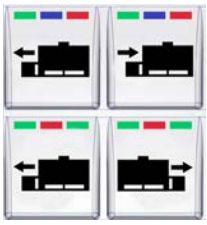


The screed operator can press the Auger Bypass button on the screed control box to pause auger operation while paving.



The screed operator can press the Auger/Conveyor Reverse button on the screed control box for reversing the direction of both functions to help clear material as needed.

NOTE: The auger/conveyor reverse function will only work when operating in the manual mode.



The screed operator can press and hold the Tunnel In or Out buttons to extend or retract the auger on the screed control boxes when paving wider lengths.

If operating in MANUAL mode:

- Press and hold the Tunnel Out button once to extend the tunnel. All LED lights illuminate green.
- Press and hold the Tunnel In button to retract the tunnel. All LED lights illuminate green.

If operating in AUTO mode:

- ***Simultaneously*** press and hold the extension control switch in the extend position and the Tunnel Out button to extend the tunnel.
- ***Simultaneously*** press and hold the extension control switch in the extend position and the Tunnel In button to retract the tunnel.
- To lock the tunnel in place, press and hold the Tunnel In and Tunnel Out buttons simultaneously until the LED light flashes green.
- To unlock, press and hold both buttons again.

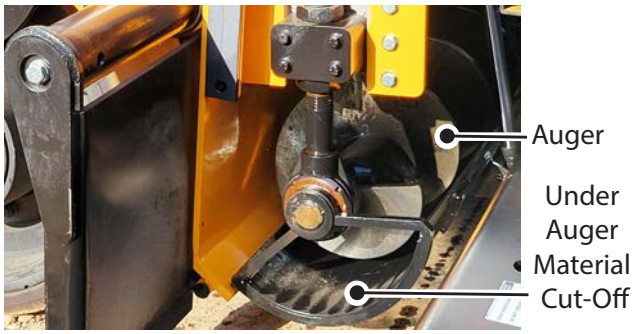


Figure 4-61. Under Auger Material Cut-Offs

Under Auger Material Cut-Offs

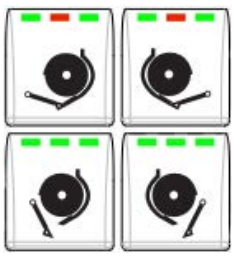
The asphalt material moves from the auger to the Under Auger Material Cut-Offs, which control asphalt flow to the screed and extensions. **(Figure 4-61)** Cut-offs open to increase asphalt flow and close to decrease flow when making narrow passes at the beginning and end of each pass or pull. Proper use of cut-offs prevents spillage and minimizes clean-up.

NOTE: To prevent excessive handwork, turn off the conveyors and close the cut-offs about three feet from the end of a pull to prevent excessive handwork.

⚠CAUTION NEVER back up with the cut-offs open to prevent machine damage.

The cut-offs can be configured to automatically close when moving paver in reverse by navigating to the Operator Settings screen on the DM430 digital controller. ***(Page 4-39)***

The Under Auger Material Cut-Offs can be operated via the main control panel and the corresponding active side on the screed control box keypad.



- Press and hold the left or right Close Cut-Off button to close.
- Press and hold the left or right Open Cut-Off button to open.

Heating the Screed, Extensions and End Gates

Heating the left and right main screed, extensions and end gates helps prevent hot mix from sticking to the cold screed plates to produce a smooth, tight mat surface. Heat the screed before paving and when the paver is idle long enough for the screed to cool between loads using the electric heat control box above the main screed.

Both the standard LB5 and optional LB7 screed heat systems are heated by a hydraulically-driven generator that powers electric heating elements. Blade heating elements in the main screed and extensions cycle during heat-up to save horsepower and fuel consumption while paving.

Both sides of the main screed elements heat when the paver is in neutral. When you start to pave, the right side heat turns off for 128 seconds while the left remains on, then turns back on, then the left sides turns off for 128 seconds. This "flip flop" system helps take some of the load off the generator for increased efficiency.

The standard LB5 Screed electric heat box features On/Off control switches for the main screed, screed extensions, heated end gates (if equipped with this option), and optional power crown control (if equipped with this option). **(Figure 4-62)** ***(See Pages 3-30 to 3-31)***



Figure 4-62. Standard LB5 Screed Electric Heat Control Box

The optional LB7 Screed electric heat box also features On/Off control switches for the main screed, screed extensions, heated end gates and [optional] power crown control. It is also equipped with a breaker switch to protect circuits from damage by overloads or short circuits. **(Figure 4-63)** ***(See Page 3-32)***



Figure 4-63. Optional LB7 Screed Electric Heat Control Box

NOTE: Lower the screed two inches off the ground while heating to prevent cold air and wind from affecting the heating process.

NOTE: Allow the screed components to heat 15 minutes before beginning to pave.

NOTE: You can set the screed directly onto a fresh mat of hot asphalt while running the heating system, allowing the heat of the asphalt to help heat the screed plate.

NOTE: Engine speed automatically increases to 1800 RPMs when the heat cycle is active.

NOTE: The engine speed automatically decreases to idle after the heat cycle completes.

NOTICE **Overheating the screed can warp the screed plate and possibly cause damage. If screed extensions lock, allow the machine to cool before attempting to extend or retract. Replace a warped screed plate immediately.**

Heat cycles for both screeds can be controlled by automatic time and temperature settings via the DM430 digital display on the main control panel or operated manually as described below.

To operate the screed heating system:

1. Start the paver and idle until the engine reaches normal operating temperature.
2. Raise the screed plate two inches off the ground.
3. Turn the desired components for heating on by moving the electric heat box switches up.



4. Press the Generator button on the control panel to activate the heating process.
5. Navigate to the Pave screen in the DM430 digital controller to set desired heating temperature and duration if operating in AUTO mode.

Once the heating function has been enabled, the electric heat box will apply electrical power to the heating elements and the heating cycle will begin. The heating cycle is timed to optimize the heat generated at the screed plates.

NOTE: The idle throttle speed ceases to function when the generator is on. When the generator turns off, the idle functionality returns.

NOTE: The generator is set at the factory to default at 1800 RPMs.

NOTE: If equipped with the power crown option, push and hold the power crown switch UP for increased crown or DOWN for increased valley.



LB5 and LB7 Screeds

The 8680 Commercial Paver features the new LB5 screed, or choose the optional LB7 screed for your paver. Both provide variable paving widths from 8 to 15.6 feet, and include berm/slope capability, horizontal depth screws, and spring-loaded end gates for matching pavement joints and seams.

Multiple adjustment mechanisms are located on top of the screed, extensions and end gate to ensure a quality mat. Electric dual vibrators mounted on the main screed frame increase paving material compaction. The optional LB7 screed includes electronic vertical height and angle-of-attack (AOA) adjustments, along with hydraulic crown/invert adjustment.

Screed operators can manage numerous functions using the screed control boxes on each side of the main screed. The screed operator can control augers/conveyors, under auger material cut-offs, tow points, and slope/grade/berm; increase/decrease Angle-of-Attack (AOA) and extension vertical; and extend/retract the screed extensions.

Screed Locks

This machine is provided with screed lock mechanisms as added safety and stability during transport. A locking lever is located on each side of the screed on the paver frame. **(Figure 4-64)**

- Push the locking lever UP to lock screed.
- Push the locking lever DOWN to unlock screed.

⚠CAUTION Ensure locks are in the unlocked position before extending or retracting the screed.

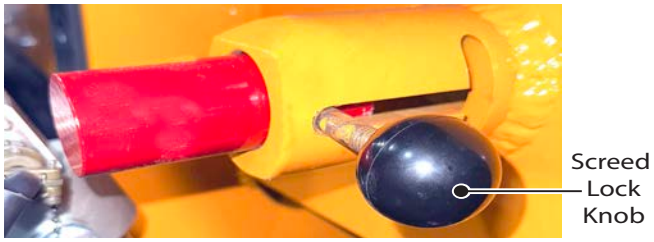


Figure 4-64. Screed Lock (Left Side Shown)

Use the procedures on the following pages for operating the screed and screed extensions:

- Always heat the screed extension(s) and end gate(s) described on the previous page before making any adjustments.
- Set the screed extension(s) and adjustment components when paving widths exceeding eight (8) feet. When adjusted correctly, pressure on the rear edge of the extended screed is the same as the rear edge of the main screed, resulting in a smooth mat the length of the screed.

NOTE: To set the asphalt depth, you can place small starter blocks under the main screed while you make adjustments. (Figure 4-65)

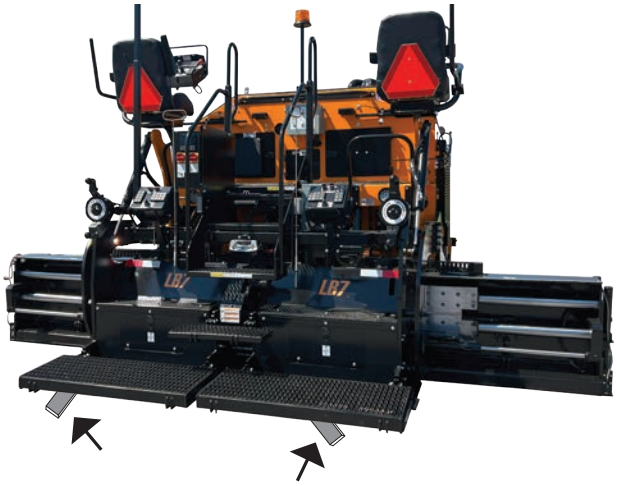


Figure 4-65. Starter Blocks Location

Use the end gate height adjusters to raise and lower the end gate height as needed. **(Figure 4-66)**

NOTE: Never allow the end gate to carry the weight of the screed to prevent uneven compaction.

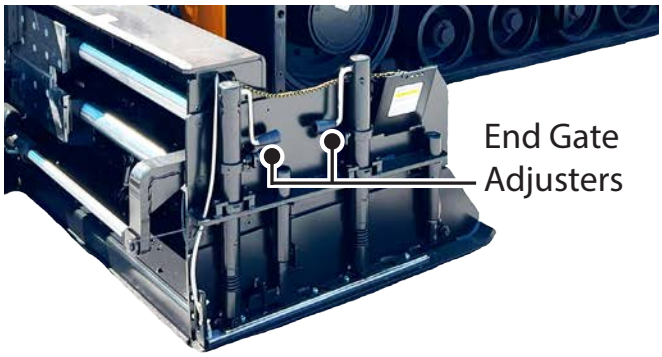


Figure 4-66. End Gate Height Adjusters

Raise/Lower Screed

Raising and lowering the screed can only be accomplished from the activated main control panel.



- Press and hold the corresponding button to raise (⇧) or lower (⇩) the screed.

- The screed lower (⇩) button also controls floating the screed:
  - Placing the joysticks into neutral (activated main control panel only) moves into screed hold, and float resumes when joysticks are returned to the forward position.
  - If the paver is moving forward, then stops, screed float is paused (the center LED light illuminates when paused. Screed float resumes when joysticks are returned to the forward position.

4